Infosys Springboard Virtual Internship 7.0 Completion Report

Team Details

Batch Number: 1

Group – 2

Team A

Start Date: June 19, 2026

Names:

S.No.	Name
1	A Jeevan Reddy
2	Asangari Guru Naga Lakshmi
3	Prathipati Sathwika
4	Uday Soni

Internship Duration: 8 Weeks

1. Project Title:

Smart urban mobility and traffic Intelligence dashboard

2. Project Objective

Urban mobility systems generate massive volumes of data across public transit, ride-hailing services, cycling networks, pedestrian infrastructure, and environmental monitoring systems. However, these datasets are often managed independently, limiting the ability of city planners and transport agencies to understand how mobility patterns interact across transportation modes. Urban Pulse is a smart city mobility intelligence platform that integrates multimodal transportation data, weather information, demographic indicators, and geospatial datasets into a unified analytics environment. The platform enables stakeholders to monitor transportation demand, analyze transit performance, evaluate mobility equity, identify infrastructure bottlenecks, and forecast future mobility trends through interactive dashboards and visual analytics. The project provides a comprehensive decision-support system that helps city administrators, transportation operators, and urban planners make data-driven decisions for sustainable and efficient urban mobility management.

3. Project Description in detail

The platform processes large-scale, multi-domain datasets to provide strategic, tactical, and operational intelligence across urban mobility networks:

- Transportation Data: Tracks multi-modal public and shared transport options (Metro, Bus, E-Bikes, E-Scooters, Bicycles), schedule adherence rates, transit connection targets, and CO_2 emissions.
- Ride-Hailing Data: Captures micro-level trip events across Cabs, Autos, and Bike Taxis, including metrics like TripID, FareAmount, SurgeMultiplier, WaitTimeMin, Cancellation Rate, DriverRating, RiderRating, and PaymentMethod (UPI, Card, Cash).
- Weather Data: Categorizes environmental conditions into discrete states (Sunny, Cloudy, Rain, Heavy Rain, Fog, Thunderstorm) to evaluate operational delays and weather-induced trip cancellations.
- Demographic Data: Classifies commuter profiles into structured personas (Students, Working Professionals, Senior Citizens, Tourists) to analyze trip distance, duration, and usage patterns.
- Geographic Data: Utilizes spatial zoning keys (ZoneKey, CityKey) and origin-destination trip vectors (DistanceKM) to identify spatial demand clusters and route friction.

Module Breakdown Across Dashboards

1. **Ride Hailing (Volume & Revenue Focus):** Summarizes overall trip trends, spatial pin mapping across India, and revenue by vehicle type, payment breakdowns, and weekly trip volume matrices.
2. **Demand & Reliability Focus (Screen shot 2 – Continuation of the raid hailing):** Analyzes of emission trends by transit availability, weather-induced trip cancellations (highest in Sunny/Cloudy conditions), surge scatter plots, and time-slot supply vs. demand gap spikes.

3. **Weather (Friction & Modal Shift Focus):** Explores trip duration vs. distance scatter distributions under weather friction, revenue degradation by weather severity, Sankey time-slot modal shifts, regional savings area charts, and 100% stacked vehicle selection charts.
4. **Micro Mobility Board (Last-Mile & Transit Integration Focus):** Focuses on micro-mobility trips by region (East, Central, North, West, South), time-slot user segment breakdowns, peak hour distribution and vehicle target performance metrics.
5. **Equity Dashboard (Accessibility & Inclusion Focus):** Analyzes high-demand rebalancing zones by city (led by Delhi, Mumbai, and Bengaluru), service gap scatter plots, state-level user adoption by weather, fare burden ratio, mobility access index, and weather target benchmarks.

4. Timeline Overview

Weeks	Milestone Number	Topics Covered
1 - 2	1	• All datasets successfully integrated and standardized. • Star schema implemented with validated relationships. • Data quality issues below defined thresholds.
3 - 4	2	• Transit reliability metrics calculated accurately. • Geospatial dashboards support multi-level drill-down. • Demand heatmaps and mobility visualizations operational.
5 - 6	3	• Ride-hailing and functioning correctly. • Modal substitution analysis produces measurable insights. • Forecasting and trend calculations validated.
7 - 8	4	• Equity metrics accurately identify accessibility gaps. • Executive dashboards provide actionable KPIs. • Complete platform refreshes within performance targets. • Final solution supports end-to-end mobility intelligence workflows.

5. Key Milestones

Weeks	Milestone Number	Topics Covered
1-2	1	Data Integration and Data Modeling
3-4	2	Geospatial and Transit Analysis

5-6	3	Demand Intelligence and Mobility Analytics
7-8	4	Equity, forecasting & Finalization

6. Project Execution Details

➢ **Data Architecture & Schema Design:** Built a centralized dimensional star-schema modeling framework connecting trip event tables (FactTrips) with dimension tables (DimWeather, DimUserType, DimVehicle, DimDate, DimGeography).

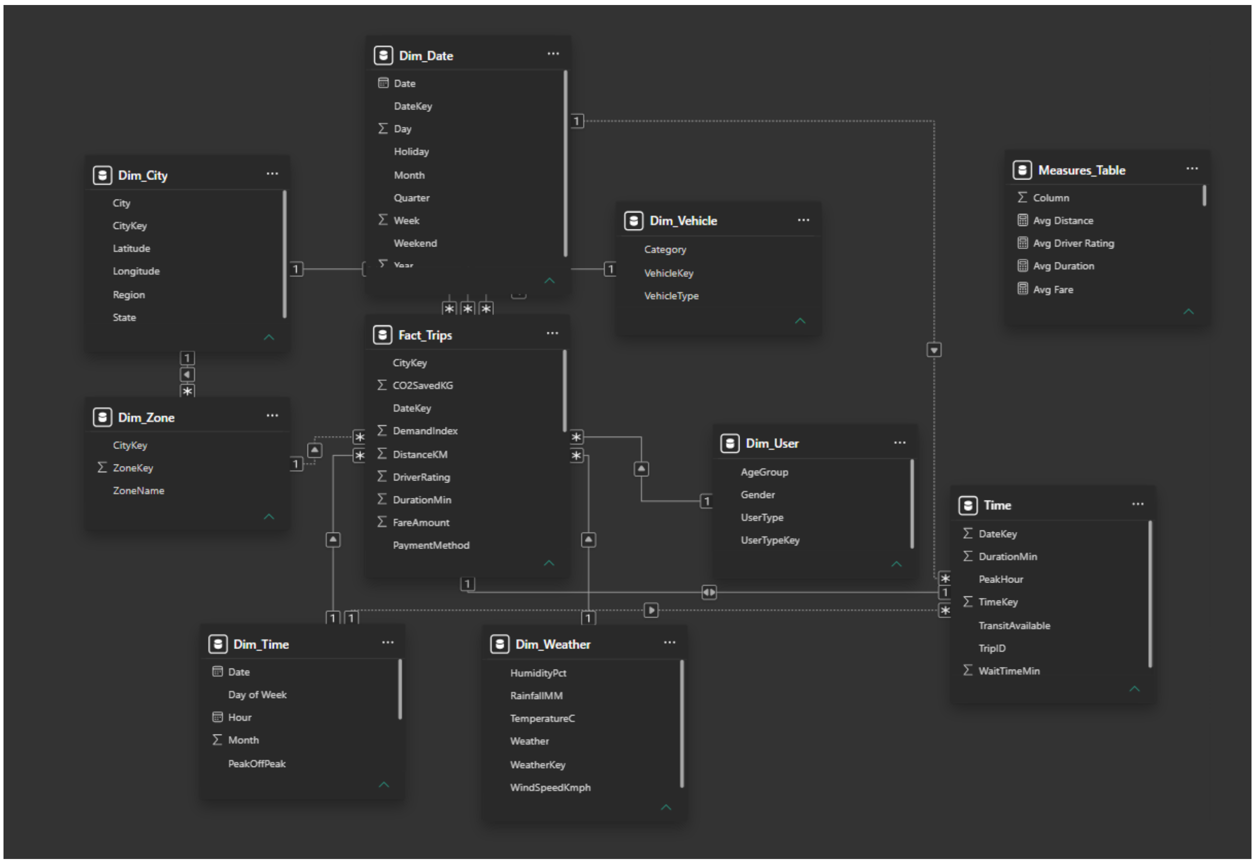


Fig1. Schema Diagram from the power BI

➢ **ETL & Data Transformation Pipeline:** Standardized raw temporal data, normalized missing spatial attributes, and feature-engineered analytical measures including demand indices, surge ratios, and weather friction scores.

Fig2. Overall Project Flow Diagram (Architecture Diagram)

- **Data Analysis & Aggregation:** Aggregated multi-source KPIs using DAX/SQL queries to evaluate route efficiency, fleet revenue contribution, and supply-demand imbalances across time slots (Morning, Afternoon, Evening, Night).

DAX Measures Used:

- Avg Distance = AVERAGE('Fact_Trips'[DistanceKM])
- Avg Driver Rating = AVERAGE('Fact_Trips'[RiderRating])
- Avg Duration = AVERAGE('Fact_Trips'[DurationMin])
- Avg Fare = AVERAGE('Fact_Trips'[FareAmount])
- Avg Wait Time = AVERAGE('Fact_Trips'[WaitTimeMin])
- Bad Weather Ratio Target = 0.15
- Bad Weather Surge Target = 1.25
- bullet target = 80.0
- Cancellation Rate =

  DIVIDE(

  CALCULATE(

  COUNTROWS('Fact_Trips'),

  TRIM('Fact_Trips'[TripStatus]) = "Cancelled"

  ),COUNTROWS('Fact_Trips'),0)

- Fare Burden Ratio =

  DIVIDE([Avg Fare], AVERAGE('Fact_Trips'[DistanceKM]), 0)
- Micromobility Trips =

  CALCULATE(

  [Total Trips],'Dim_Vehicle'[VehicleType] IN {"E-Bike", "E-Scooter", "Bicycle"})
- Mobility Access Index =

  VAR TransitScore = DIVIDE([Transit Connected Trips], [Total Trips], 0) * 40

  VAR WaitTimeScore = MAX(0, 30 - (AVERAGE('Fact_Trips'[WaitTimeMin]) * 1.5))

  VAR SupplyScore = MIN(30, AVERAGE('Fact_Trips'[SupplyIndex]) * 0.3)

  RETURN

  ROUND(TransitScore + WaitTimeScore + SupplyScore, 1)
- Mobility Access Index 1 =

  VAR TransitScore = DIVIDE(CALCULATE([Total Trips], 'fact_trips'[TransitAvailable] = "Yes"), [Total Trips], 0) * 100

  VAR WaitScore = MAX(0, 100 - (AVERAGE('fact_trips'[WaitTimeMin]) * 4))

  VAR FareScore = MAX(0, 100 - (AVERAGE('fact_trips'[FareAmount]) * 0.5))

  RETURN

  (TransitScore * 0.40) + (WaitScore * 0.30) + (FareScore * 0.30)
- Peak Hour Trips =

  CALCULATE(

  [Total Trips],

  'Fact_Trips'[PeakHour] = "Yes" || 'Fact_Trips'[PeakHour] = "TRUE")
- Public Transit Trips =

  CALCULATE(

  [Total Trips],

  'Dim_Vehicle'[VehicleType] IN {"Metro", "Bus"})
- Revenue Target =

  CALCULATE(

  [Total Revenue],

  PREVIOUSMONTH('Dim_Date'[Date])) * 1.08
- Ride Hailing Trips =

  CALCULATE(

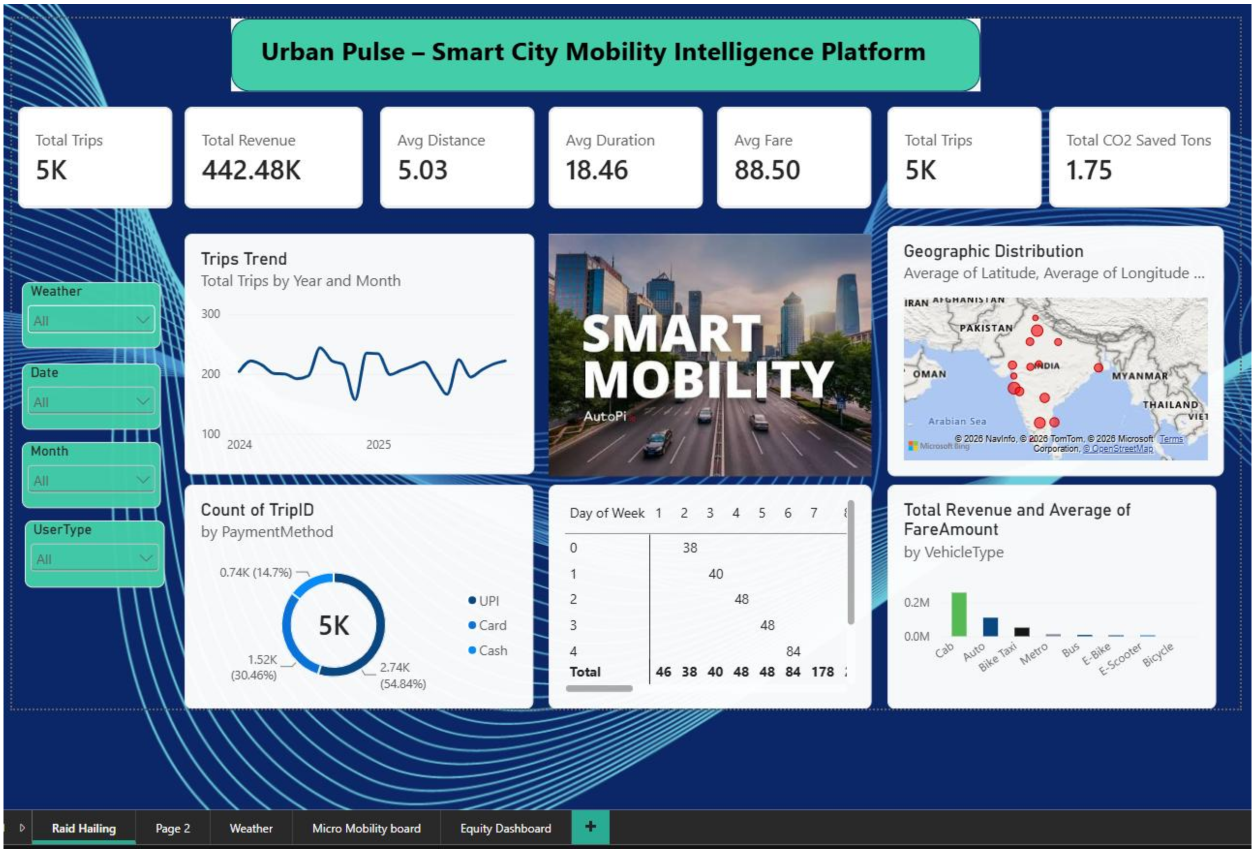
  [Total Trips],

  'Dim_Vehicle'[VehicleType] IN {"Cab", "Auto", "Bike Taxi"})

- Schedule Adherence Rate =
  DIVIDE(
  CALCULATE(COUNTROWS('Fact_Trips'), 'Fact_Trips'[WaitTimeMin] <= 5),
  COUNTROWS('Fact_Trips'),0)
- Target Driver Rating = 5
- Total CO2 Saved Tons = DIVIDE(SUM('Fact_Trips'[CO2SavedKG]), 1000, 0)
- Total Revenue = SUM('Fact_Trips'[FareAmount])
- Total Trips = COUNTROWS('Fact_Trips')
- Transit Connected Trips =
  CALCULATE(
  [Total Trips],
  'Fact_Trips'[TransitAvailable] = "Yes" || 'Fact_Trips'[TransitAvailable] = "TRUE")
- Transit Connection Target = 0.40
- Transit Desert Status =
  IF(
  AVERAGE('Fact_Trips'[DemandIndex]) > 60 && [Transit Connected Trips] / [Total Trips] <
  0.20,
  "Transit Desert",
  "Adequate Service")
- Wait Time Target = 8.0

- **Interactive Dashboard Development:** Designed five target-driven Power BI report views focusing on Operational Volume, Demand & Weather Reliability, User Behavior & Modal Shift, and Financial/Target Performance.

7. Screenshots/Snapshots


Urban Pulse – Smart City Mobility Intelligence Platform
Total Trips
5K
Total Revenue
442.48K
Avg Distance
5.03
Avg Duration
18.46
Avg Fare
88.50
Total Trips
5K
Total CO2 Saved Tons
1.75
Weather
All
Date
All
Month
All
UserType
All
Trips Trend
Total Trips by Year and Month
300
200
100
2024
2025
SMART MOBILITY
Geographic Distribution
Average of Latitude, Average of Longitude ...
Count of TripID
by PaymentMethod
0.74K (14.7%)
5K
1.52K (30.46%)
2.74K (54.84%)
UPI
Card
Cash
Day of Week 1 2 3 4 5 6 7
0 38
1 40
2 48
3 48
4 84
Total 46 38 40 48 48 84 178
Total Revenue and Average of FareAmount
by VehicleType
0.2M
0.0M
Cab
Auto
Bike Taxi
Metro
Bus
E-Bike
E-Scooter
Bicycle
Raid Hailing
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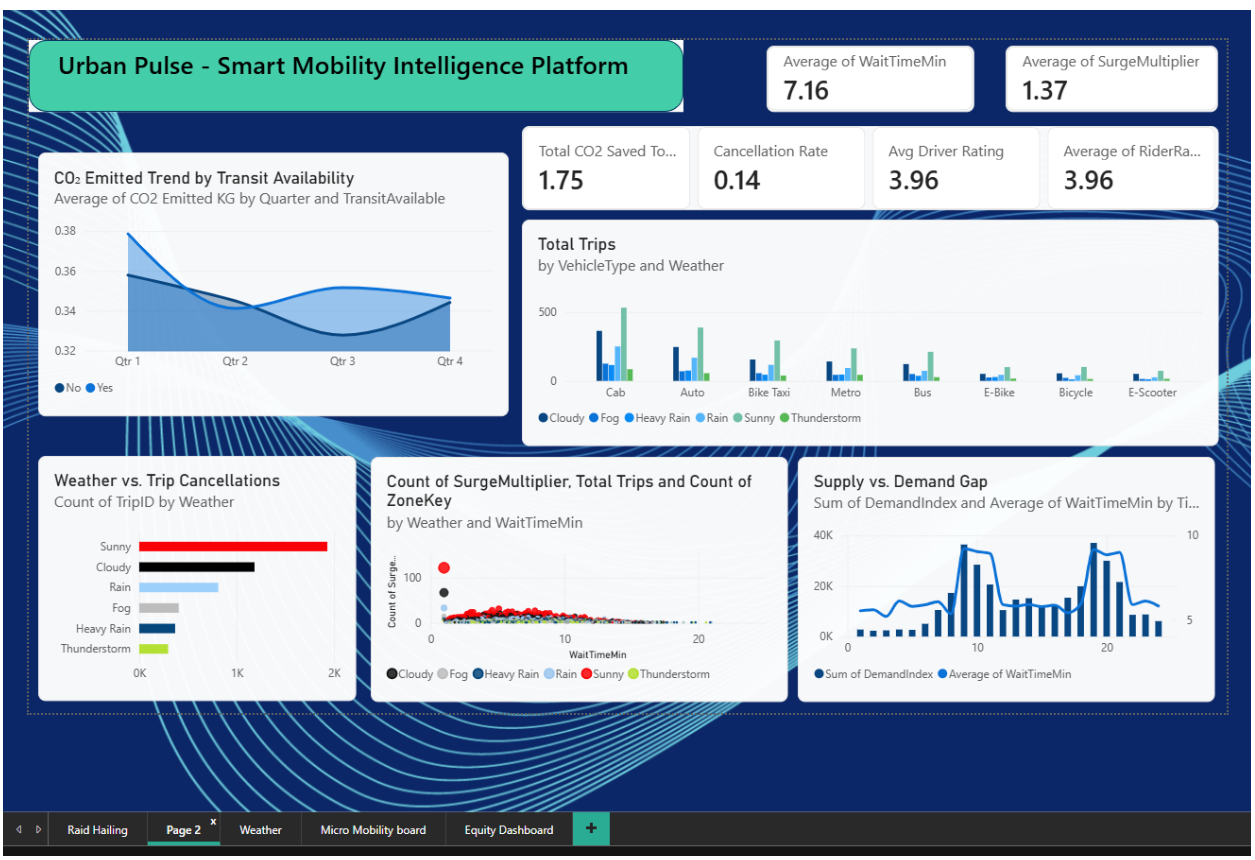
Weather
Micro Mobility board
Equity Dashboard


Urban Pulse - Smart Mobility Intelligence Platform
Average of WaitTimeMin
7.16
Average of SurgeMultiplier
1.37
Total CO2 Saved To...
1.75
Cancellation Rate
0.14
Avg Driver Rating
3.96
Average of RiderRa...
3.96
CO₂ Emitted Trend by Transit Availability
Average of CO2 Emitted KG by Quarter and TransitAvailable
0.38
0.36
0.34
0.32
Qtr 1
Qtr 2
Qtr 3
Qtr 4
No
Yes
Total Trips
by VehicleType and Weather
500
0
Cab
Auto
Bike Taxi
Metro
Bus
E-Bike
Bicycle
E-Scooter
Cloudy
Fog
Heavy Rain
Rain
Sunny
Thunderstorm
Weather vs. Trip Cancellations
Count of TripID by Weather
Sunny
Cloudy
Rain
Fog
Heavy Rain
Thunderstorm
0K
1K
2K
Count of SurgeMultiplier, Total Trips and Count of ZoneKey
by Weather and WaitTimeMin
100
0
0
10
20
WaitTimeMin
Cloudy
Fog
Heavy Rain
Rain
Sunny
Thunderstorm
Supply vs. Demand Gap
Sum of DemandIndex and Average of WaitTimeMin by Ti...
40K
20K
0K
0
10
20
10
5
Sum of DemandIndex
Average of WaitTimeMin
Raid Hailing
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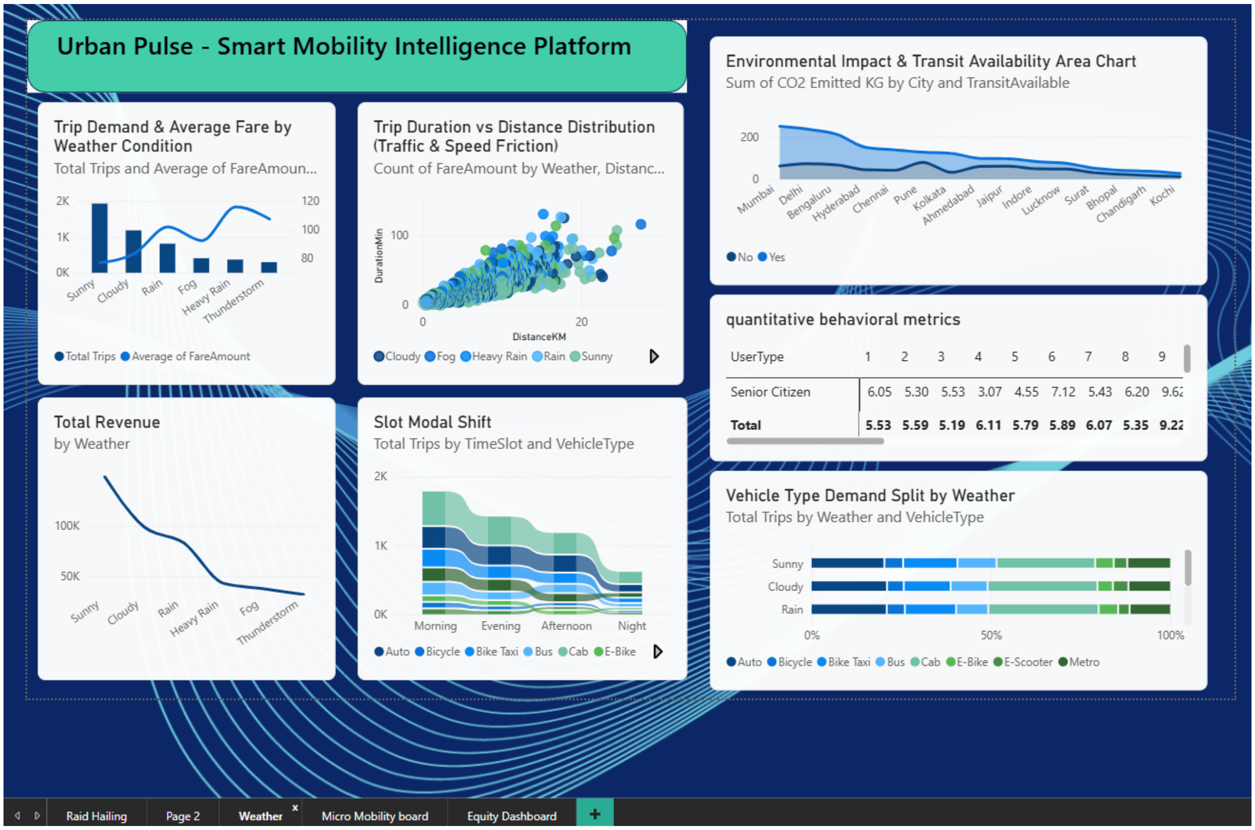
Weather
Micro Mobility board
Equity Dashboard


Urban Pulse - Smart Mobility Intelligence Platform
Trip Demand & Average Fare by Weather Condition
Total Trips and Average of FareAmoun...
Trip Duration vs Distance Distribution (Traffic & Speed Friction)
Count of FareAmount by Weather, Distanc...
Environmental Impact & Transit Availability Area Chart
Sum of CO2 Emitted KG by City and TransitAvailable
quantitative behavioral metrics
UserType 1 2 3 4 5 6 7 8 9
Senior Citizen 6.05 5.30 5.53 3.07 4.55 7.12 5.43 6.20 9.62
Total 5.53 5.59 5.19 6.11 5.79 5.89 6.07 5.35 9.22
Total Revenue
by Weather
Slot Modal Shift
Total Trips by TimeSlot and VehicleType
Vehicle Type Demand Split by Weather
Total Trips by Weather and VehicleType
Raid Hailing
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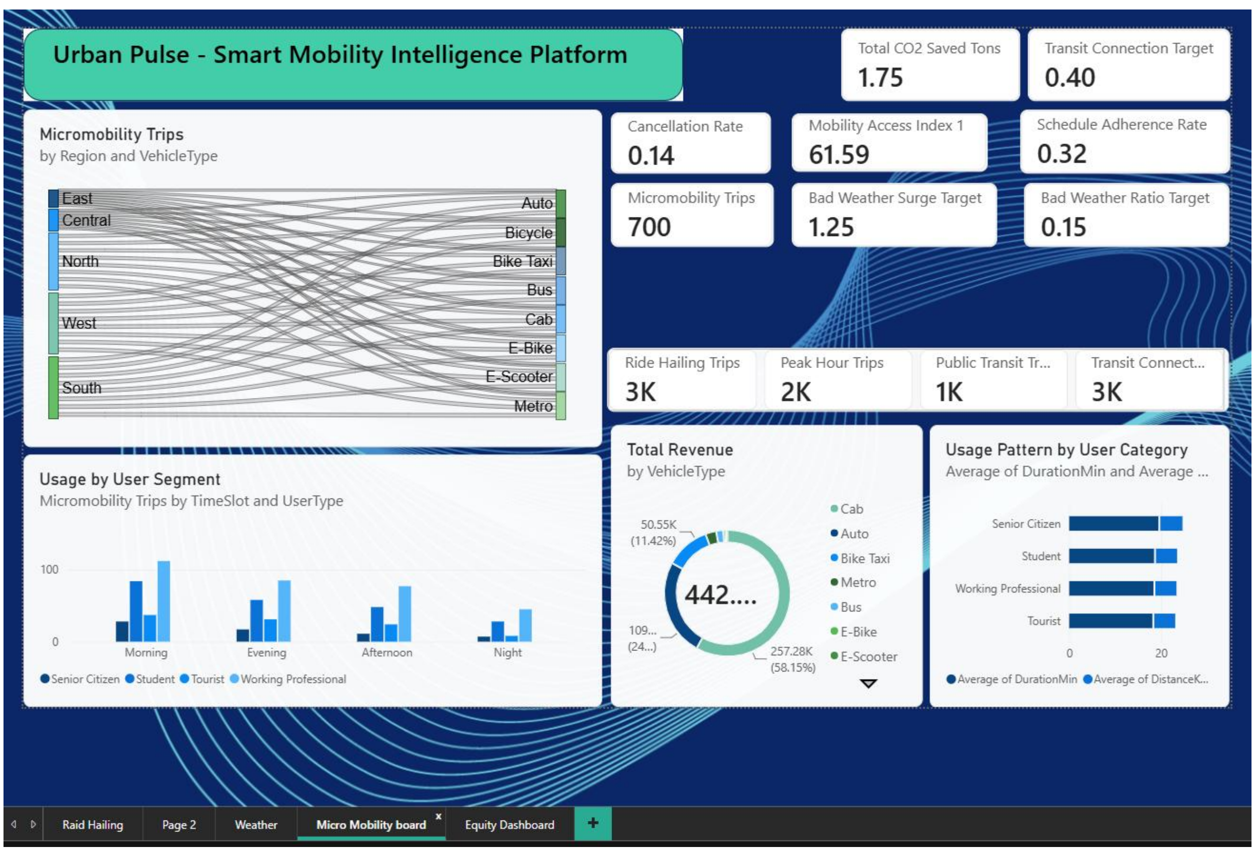
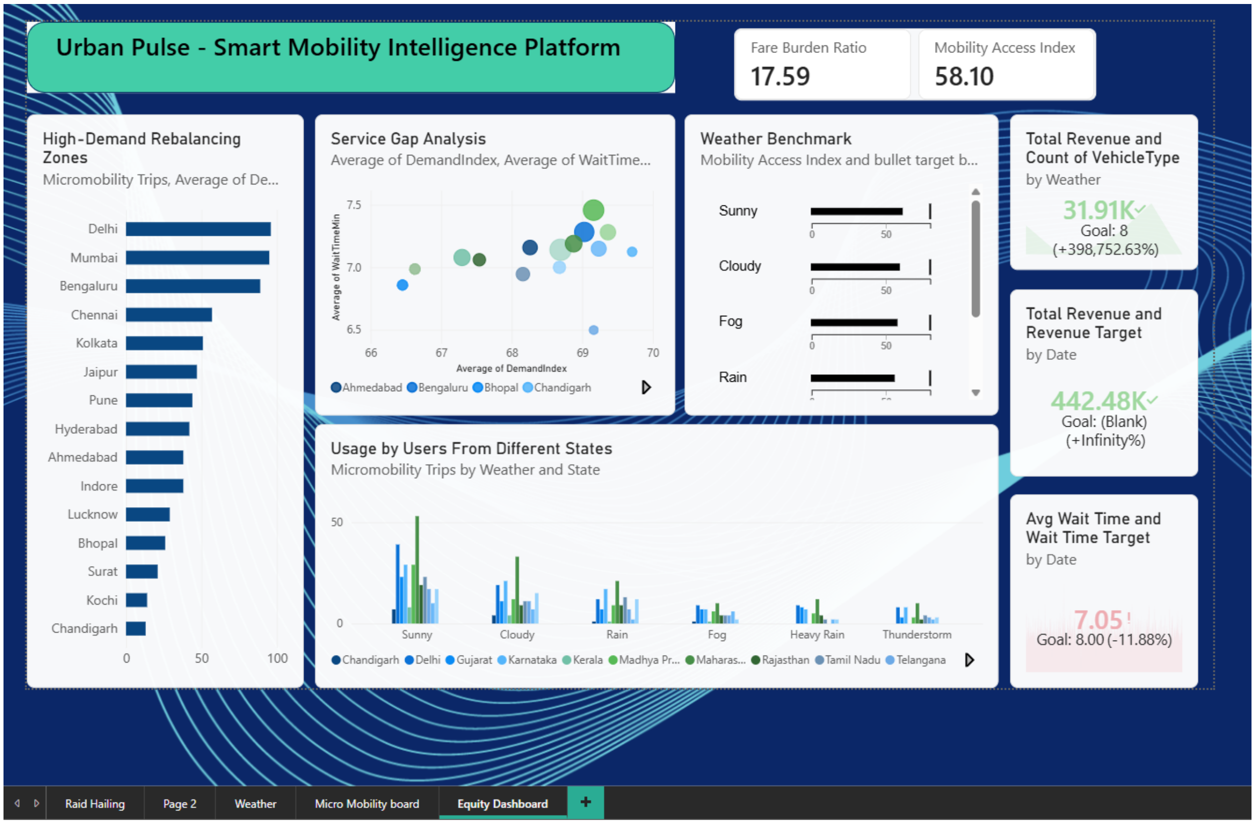
Weather
Micro Mobility board
Equity Dashboard


Urban Pulse - Smart Mobility Intelligence Platform
Total CO2 Saved Tons
1.75
Transit Connection Target
0.40
Cancellation Rate
0.14
Mobility Access Index 1
61.59
Schedule Adherence Rate
0.32
Micromobility Trips
700
Bad Weather Surge Target
1.25
Bad Weather Ratio Target
0.15
Micromobility Trips
by Region and VehicleType
Ride Hailing Trips
3K
Peak Hour Trips
2K
Public Transit Tr...
1K
Transit Connect...
3K
Usage by User Segment
Micromobility Trips by TimeSlot and UserType
Total Revenue
by VehicleType
442....
Usage Pattern by User Category
Average of DurationMin and Average ...
Raid Hailing
Page 2
Weather
Micro Mobility board
Equity Dashboard



8. Challenges Face

- **Multi-Source Data Integration:** Harmonizing fine-grained ride-hailing event logs with broader public transit timetables and weather timestamps required strict ETL temporal alignment.
- **Spatial Visual Mapping Restrictions:** Standard map visual rendering faced tenant configuration policies in the reporting environment, requiring alternative matrix, key-based, and scatter plot spatial representations.
- **Data Sparsity under Severe Weather Events:** Balancing extreme surge spikes and low sample volumes during rare weather events (e.g., severe thunderstorms) without skewing baseline statistical averages.

9. Learning’s and Skills Acquired

- **Business Intelligence & Visualization:** Enterprise dashboard layout design, data storytelling, custom color palettes, KPI card design, and UI/UX optimization in Power BI.
- **Data Modeling & Analytics:** Star-schema dimensional modeling, writing complex DAX metrics, calculating running totals, and performing gap analysis.
- **Domain Knowledge:** Operational understanding of urban mobility dynamics, surge pricing algorithms, modal shift behavior, and smart city environmental metrics.
- **Tools:** Excel, Power BI, SQL, Tableau

10. Testimonials from Team

"Designing the data pipelines for Urban Pulse allowed us to see how weather patterns immediately impact commuter behavior. Translating complex spatial data into actionable dashboard visuals was an immensely rewarding experience."

"Developing the modal shift and financial goal-tracking components provided deep insights into how multi-modal integration can reduce city transit bottlenecks and carbon emissions."

11. Conclusion

The **Urban Pulse Smart City Mobility Intelligence Platform** demonstrates the power of unifying multi-domain urban datasets into a cohesive decision-support system. By linking ride-hailing metrics with public transit accessibility, commuter demographics, and weather severity, the platform offers actionable insights for urban planning, fleet optimization, and operational risk mitigation.

12. Acknowledgement

We express our gratitude to our academic mentor (Dr. P Satya), domain advisors, team mates and peer collaborators who provided continuous guidance, technical insights, and evaluation feedback throughout the design, modeling, and dashboard development phases of this project.